ORF307: Optimization

Precept: #2

Agenda

- Ballistic example
- Inverse of a matrix with LU factorization
- More plotting examples (see notebooks in Companion)

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW: ....
- Other announcements on blackboard.

Ballistic example

Overview:

- a projectile is moving in a 2-dim space
- we sample its position and velocity at times $\tau = 0, 1h, 2h, \ldots, Th$ (so we sample $T + 1$ times)
- $p_t \in \mathbb{R}^2$ is the position at time $\tau = th$
- $v_t \in \mathbb{R}^2$ is the velocity at time $\tau = th$
- $f_t \in \mathbb{R}^2$ is the total force at time $\tau = th$
- $x_t = (p_t, v_t)^T \in \mathbb{R}^4$ is the projectile's state at time $\tau = th$

Formal model:

- $f_t = mg - \eta v_t$ (this is a common approximation for a projectile experiencing a constant gravitational force and a drag force proportional to its velocity).
- $\eta \in \mathbb{R}$ is the drag coefficient (note that in f_t, the term ηv_t has a negative sign, because it acts in the opposite direction of the velocity).

- $g = (0, -9.8)^T$ is gravity

Dynamics:

- approximate the velocity as constant over the time interval $th \leq \tau \leq (t+1)h$

- approximate force as constant over the same time interval

Now we can write this more compactly in matrix-vector form as

$$x_{t+1} = Ax_t + b$$

and define the matrix A and vector b

Propagating the state through time:

Targeting problem

Given:

- initial position p_0
- parameters h, m, η
- flight time : Th
- desired final position (target): p_T

Goal:

- find the initial velocity v_0

We will begin by determining the final state x_T ......

Robust ballistics

- Suppose we have uncertainty in the drag coefficient
- Uncertainty can be modeled as K scenarios
- Each scenario k has its own $A^{(k)}, b^{(k)}, C^{(k)}, d^{(k)}$

Robust targeting

How can you formulate this as an optimization problem?